

# Soumik Dutta Choudhury

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## PROFESSIONAL EXPERIENCE

### Freelance Software Developer

2023 – Present

- Delivered end-to-end web and machine learning projects, managing frontend, backend, and deployment workflows.
- Built responsive and user-focused web applications using **React, Django, and Python**.
- Implemented clean APIs, authentication, and database integration for real-world use cases.
- Deployed projects on cloud platforms with proper version control using **Git**.
- Worked closely with clients to convert requirements into functional, scalable solutions

## EDUCATION

### Bachelor of Technology (B.Tech) in Computer Science & Engineering — CGPA: 6.2

2022 – 2026

Central Institute of Technology Kokrajhar, Assam

## TECHNICAL SKILLS

### Languages

Python, C, C++, Java, SQL, JavaScript, HTML/CSS, MySQL

### Frameworks/Libraries

HTML, CSS, JavaScript, React, Flask, Django, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib

### Developer Tools

Git, GitHub, VS Code, Jupyter Notebook, Linux, Auth0, Docker (Basics)

### CS Fundamentals

Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), OS, DBMS

## PROJECT EXPERIENCE

### Deepfake Detection System using Vision Transformer (ViT)

Python, TensorFlow

- Developing an advanced computer vision system using ViT to identify manipulated facial imagery with high precision.
- Implementing Grad-CAM++ for model explainability, allowing for visual verification of detection heatmaps.
- Building modular data pipelines for preprocessing, augmentation, and evaluation of large-scale video datasets.

### Brahmi to Hindi OCR and Translation System

CNN, OpenCV, Python

- Engineered an end-to-end pipeline to preserve ancient scripts by converting Brahmi characters into modern Hindi.
- Trained a CNN-based classifier on 850+ specialized samples, achieving high recognition accuracy in noisy images.
- Integrated EAST Text Detection for localized script extraction and Seq2Seq models for linguistic translation.

### Parkinson's Disease Detection via EEG Signals

Machine Learning, Signal Processing

- Developed a diagnostic ML pipeline to classify Parkinson's disease using multi-channel EEG signal data.
- Conducted rigorous feature extraction and signal denoising to improve supervised model performance and reliability.

### Bitcoin Trader Performance & Sentiment Analysis

Data Analysis, Python

- Analyzed time-series market data to correlate social sentiment with cryptocurrency trader profitability patterns.